

Offensive Security Internship Project Report

Full Name	- Raj Naarayanan
Intern Id	- ITID0757
Batch start	- 25 May 2025
Duration of months	- 6 Months
Ending date	- 25 November 2025
Collab link-GitHub link	- https://github.com/Cristy11001/PDF_Protector_python.git

https://github.com/Cristy11001/Back_door-Reverse_engineering-.git

https://github.com/Cristy11001/Basic_Network_scanning.git

https://github.com/Cristy11001/Simple_Port_scanner-Python-.git

Basic_Network_Scanning_Report:

2. Threading

- Scanning each port sequentially is slow.
- By using Python's `threading` module, multiple port checks run simultaneously.
- Each thread handles a single port or a small group of ports.
- This greatly improves performance and allows near real-time scanning of large port ranges.

Core Functional Components

1. Port Scanning Function

Attempts to connect to a specified port using a socket. If the connection succeeds, the port is marked as open.

2. Thread Manager

Creates and starts multiple threads so that many ports can be scanned at the same time.

3. Network Range or Single Host Input

The script may allow scanning:

→ a single IP

→ a list of IPs

→ a subnet (optional depending on your implementation)

4. Result Display and Logging

The script prints open ports and active hosts and can optionally save results to a file.

Summary

This basic network scanner demonstrates the essential principles of network reconnaissance in a simple and understandable manner. Using Python's built-in `socket` module allows direct interaction with network endpoints, while threading provides faster execution by scanning many ports concurrently.

The project teaches:

- how port scanning works,
- how to create TCP connections programmatically,
- how threading can optimize performance,
- how to structure modular functions for scanning tasks.

Although minimalistic, it provides a practical introduction to network analysis and security testing.

The Basic Network Scanning Project successfully shows how Python can be used to create a lightweight and efficient network scanner using only the `socket` module and threading. It highlights the importance of concurrency in network operations and forms a strong foundation for building more advanced tools, such as multi-protocol scanners, service detection, or vulnerability analysis modules. This project should be used responsibly and only on networks where permission has been granted. As a learning exercise, it serves as an excellent starting point for students and beginners in cybersecurity and networking.